

Information Field Theory

And Geometry / Gravity
IFT-Geometry & Gravity v2.0

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The relationship between Information Field Theory (IFT) and gravity is presented. The central thesis is that physical geometry is coupled to an informational geometry induced by the field $\rho(x) > 0$. The auxiliary informational metric $\tilde{g}_{\mu\nu}[\rho] = \nabla_\mu \nabla_\nu \ln \rho - \lambda_F \Xi_\mu \Xi_\nu$ is defined, and the emergent Lorentzian signature theorem is proved. Geometry–information coupling is introduced via S_{align} , and General Relativity is recovered as a limit when $\nabla u \rightarrow 0$. Corrections to gravitational waves are derived: $h(f) = h_{\text{GR}}(f)[1 + \kappa_1(f/f_{\text{ref}})]$, with $\kappa_1 \sim 10^{-15}$ to 10^{-14} . Current LIGO/Virgo bounds ($\kappa_1 < 4.9 \times 10^{-14}$) and projections for the Einstein Telescope and Cosmic Explorer are analyzed. Informational black holes, basic geometric cosmology, and falsification criteria are discussed.

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Chapter 1

Introduction: Two Geometries

1.1 IFT Core

IFT starts from a positive scalar field $\rho(x) > 0$. The derived variables are:

$$u = \sqrt{\rho}, \quad \varphi = \ln \rho, \quad \Xi_\mu = \nabla_\mu \ln \rho, \quad \Xi = |\nabla \ln \rho|. \quad (1.1)$$

1.2 Central Thesis

IFT-Geometry rigorously distinguishes two geometric objects:

- $g_{\mu\nu}$: physical metric, a dynamic tensor field describing gravitation.
- $\tilde{g}_{\mu\nu}[\rho]$: auxiliary informational metric, derived from the field ρ .

The prudent thesis is:

Physical geometry can be coupled to an informational geometry derived from $\rho(x)$.

 (1.2)

The strong (still conjectural) thesis is:

Spacetime can emerge as an effective structure from the relational information $\rho(x)$.

 (1.3)

Chapter 2

Informational Geometry

2.1 Local Fisher Tensor

$$I_{\mu\nu}(x) = \Xi_\mu(x) \Xi_\nu(x) = \partial_\mu \varphi \partial_\nu \varphi. \quad (2.1)$$

Symmetric, rank-1, positive semidefinite.

2.2 Dilaton Hessian

$$H_{\mu\nu}(x) = \nabla_\mu \nabla_\nu \varphi(x) = \nabla_\mu \nabla_\nu \ln \rho(x). \quad (2.2)$$

Contains information about the local curvature of the informational field.

2.3 Auxiliary Informational Metric

$$\boxed{\tilde{g}_{\mu\nu}[\rho] = H_{\mu\nu} - \lambda_F I_{\mu\nu} = \nabla_\mu \nabla_\nu \ln \rho - \lambda_F (\nabla_\mu \ln \rho)(\nabla_\nu \ln \rho)}, \quad (2.3)$$

with $\lambda_F > 0$ the Fisher deformation parameter.

Chapter 3

Emergent Lorentzian Signature Theorem

Theorem 3.1 (Local Lorentzian Signature). *Let $p \in M$ be a point where $H_{\mu\nu}(p)$ is positive definite and $\Xi_\mu(p) \neq 0$. Define the critical value $\lambda^* = h_{\min}/|\Xi|^2$. Then, for all $\lambda_F > \lambda^*$, the metric $\tilde{g}_{\mu\nu}$ has signature $(-, +, +, +)$ at p .*

Proof. $\tilde{g}_{\mu\nu} = H_{\mu\nu} - \lambda_F \Xi_\mu \Xi_\nu$ is a rank-1 perturbation of $H_{\mu\nu}$. By the rank-1 update lemma, the eigenvalue most aligned with Ξ is shifted downward. When $\lambda_F > \lambda^*$, exactly one eigenvalue becomes negative while the others remain positive. $\square$

Corollary 3.2 (Emergent Time Direction). *The eigenvector with the negative eigenvalue aligns with Ξ^μ , the direction of maximal informational gradient:*

$$t^\mu \approx \frac{\Xi^\mu}{|\Xi|}. \quad (3.1)$$

Interpretation: *Time emerges locally as the direction of maximal informational flow, without needing to postulate it.*

Rigor note: This is a local result. The global existence of a regular Lorentzian metric requires additional conditions on ρ (global non-degeneracy, orientability of the eigenvector field).

Chapter 4

Geometry–Information Coupling

4.1 Alignment Action

$$S_{\text{align}} = \frac{1}{2\sigma^2} \int_M d^4x \sqrt{-g} \|g_{\mu\nu} - \tilde{g}_{\mu\nu}[\rho]\|^2. \quad (4.1)$$

4.2 Regimes

- $\sigma \rightarrow \infty$: total decoupling. Standard gravity (GR).
- σ finite: intermediate coupling. Small corrections to GR.
- $\sigma \rightarrow 0$: strong emergent geometry. $g_{\mu\nu} \rightarrow \tilde{g}_{\mu\nu}[\rho]$ (conjectural).

Chapter 5

Complete Gravitational Action

5.1 Conservative Formulation

$$\boxed{S_{\text{IFT-grav}} = S_{\text{EH}} + S_{\rho} + S_{\text{align}} + S_{\text{corr}}}, \quad (5.1)$$

where:

$$S_{\text{EH}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} (R - 2\Lambda_0), \quad (5.2)$$

$$S_{\rho} = \int d^4x \sqrt{-g} \left[\frac{Z_{\rho}}{2} g^{\mu\nu} \nabla_{\mu} u \nabla_{\nu} u - V(u) \right]. \quad (5.3)$$

5.2 Modified Einstein Equations

$$G_{\mu\nu} + \Lambda_0 g_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{(\rho)} + T_{\mu\nu}^{(\text{align})} + T_{\mu\nu}^{(\text{corr})} + T_{\mu\nu}^{(\text{matter})} \right). \quad (5.4)$$

5.3 General Relativity Limit

When $\rho \rightarrow \rho_0 = \text{constant}$, $\nabla u \rightarrow 0$, $\Xi \rightarrow 0$, and $S_{\text{align}} \rightarrow 0$:

$$S_{\text{IFT-grav}} \rightarrow \frac{1}{16\pi G} \int d^4x \sqrt{-g} (R - 2\Lambda_{\text{eff}}), \quad (5.5)$$

with $\Lambda_{\text{eff}} = \Lambda_0 + 8\pi G V(v) + 8\pi G/\sigma^2$. This is GR with an effective cosmological constant.

Chapter 6

Gravitational Waves in IFT

6.1 Phase Correction

Gravitational wave propagation is modified by fluctuations of the informational field along the line of sight:

$$h(f) = h_{\text{GR}}(f) \left[1 + \sum_n \kappa_n \left(\frac{f}{f_*} \right)^n \right]. \quad (6.1)$$

The dominant correction is linear in frequency:

$$h(f) \approx h_{\text{GR}}(f) \left[1 + \kappa_1 \left(\frac{f}{f_{\text{ref}}} \right) \right]. \quad (6.2)$$

6.2 Physical Origin

Fluctuations $\delta\rho$ of the informational field along the propagation path produce an effective velocity variation, accumulating phase proportional to frequency:

$$\Delta\Phi(f) = \kappa_1 \frac{f}{f_{\text{ref}}} \int_0^L \frac{\delta\rho(x)}{\rho_0} dl. \quad (6.3)$$

6.3 Derivation of κ_1

$$\kappa_1 = \frac{\delta\rho}{\rho_0} \cdot \frac{1}{\sigma^2} \cdot \frac{c}{H_0} \cdot \frac{r_s}{\ell_P}. \quad (6.4)$$

With $\delta\rho/\rho_0 \sim 10^{-5}$ (inflationary fluctuations), $\sigma \sim M_P$, $r_s \sim \text{Gpc}$, $H_0 \sim 10^{-18} \text{ s}^{-1}$:

$$\kappa_1 \sim 10^{-15} \text{ to } 10^{-14}. \quad (6.5)$$

6.4 Observational Bounds

- **GW170817:** $|\kappa_1| < 4.9 \times 10^{-14}$ (90% confidence).
- **GWTC-3:** $\kappa_1 = 0 \pm \mathcal{O}(10^{-14})$.

The IFT prediction lies just below current bounds. It is not ruled out.

6.5 Predictions for Third-Generation Detectors

- **Einstein Telescope (ET)**: sensitivity $\Delta\kappa_1 \sim 10^{-16}$.
- **Cosmic Explorer (CE)**: sensitivity $\Delta\kappa_1 \sim 10^{-16}$.

If IFT is correct, ET/CE will detect $\kappa_1 \neq 0$ at $> 3\sigma$ in the 2030s.

Chapter 7

Informational Black Holes

7.1 Horizon as a Region of High Informational Structure

Near the horizon $r_s = 2GM/c^2$, the informational gradient diverges smoothly:

$$\Xi(r) \sim \frac{1}{r - r_s}. \quad (7.1)$$

7.2 Horizon Correction

$$r_{\text{hor}}^{\text{IFT}} = r_s + \delta r, \quad \delta r \sim \frac{Z_\rho v^2}{\sigma^2 M}. \quad (7.2)$$

For astrophysical black holes ($M \gg M_P$), $\delta r \ll r_s$, undetectable with current technology.

7.3 Bekenstein–Hawking Entropy

IFT reinterprets black hole entropy as the Shannon entropy of the informational field at the horizon:

$$S_{\text{IFT}} = -k_B \int_{\text{horizon}} \rho \ln \rho \, dA = \frac{k_B A}{4\ell_P^2} + \text{logarithmic corrections}. \quad (7.3)$$

Chapter 8

Predictions and Falsification Criteria

8.1 Secure Predictions

1. GR is exactly recovered when $\nabla u \rightarrow 0$.
2. The Lorentzian signature emerges locally from Fisher geometry.
3. The informational Friedmann equation reproduces FLRW.
4. Bekenstein–Hawking entropy is reproduced at leading order.

8.2 Testable Predictions

1. Linear correction in gravitational waves: $h(f) = h_{\text{GR}}(f)[1 + \kappa_1(f/f_{\text{ref}})]$ with $\kappa_1 \sim 10^{-15}$.
2. Dynamical cosmological equation of state: $w(z) = -1 + \varepsilon z/(1+z)$.
3. Horizon correction for black holes (currently undetectable, but conceptually defined).

8.3 Falsification Criteria

1. If $\kappa_1 = 0$ with precision $< 10^{-16}$ (ET/CE), IFT-Grav is constrained (though not refuted, since $\sigma \rightarrow \infty$ is a permitted limit).
2. If $w(z) = -1$ exactly at 5σ in DESI+Euclid+CMB-S4, the dynamical version of IFT is ruled out.
3. If violations of causality or ghosts are detected in the gravitational sector.

Chapter 9

Master Formula of IFT-Geometry & Gravity

$$\rho > 0, \quad \varphi = \ln \rho, \quad \Xi_\mu = \nabla_\mu \ln \rho, \quad (9.1)$$

$$H_{\mu\nu} = \nabla_\mu \nabla_\nu \ln \rho, \quad (9.2)$$

$$\tilde{g}_{\mu\nu}[\rho] = H_{\mu\nu} - \lambda_F \Xi_\mu \Xi_\nu, \quad (9.3)$$

$$S_{\text{IFT-grav}} = \frac{1}{16\pi G} \int \sqrt{-g} (R - 2\Lambda_0) + S_\rho + S_{\text{align}}, \quad (9.4)$$

$$h(f) = h_{\text{GR}}(f) \left[1 + \kappa_1 \left(\frac{f}{f_{\text{ref}}} \right) \right], \quad \kappa_1 \sim 10^{-15}. \quad (9.5)$$

Chapter 10

Final Declaration

IFT-Geometry proposes that physical geometry is coupled to an informational geometry induced by $\rho(x)$. The conservative version (currently the most robust) asserts that ρ defines an auxiliary geometry $\tilde{g}_{\mu\nu}$ that is couplable to the physical metric $g_{\mu\nu}$. The theory recovers GR in the limit $\nabla u \rightarrow 0$ and allows controlled corrections that are testable with third-generation gravitational-wave detectors.

IFT-gravity = GR + coupled informational geometry, with strong emergence as a conjecture .
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